EYES IN THE SKY

AI BIRD DETECTION FOR DRONE SAFETY

“PI”RONMAN

OUR TEAM & STATUS

Name	Role	Description
Jordan Capelle	Systems Integration Engineer	Hardware setup & YOLO integration
Karl Garman	Research & Analysis Engineer	Literature review & documentation
Mohomad Sabur	Lead Software Engineer	AI model coding & optimization

Hardware Setup

Code / Train (November 19th)

Test / Validate

Presentation / Paper (December 17th)

Bottom Line: On schedule with no major blockers.

LITERATURE REVIEW

None of the reviewed papers used a YOLO model on an AI chip and applied it to the use case of identifying avian flight hazards.

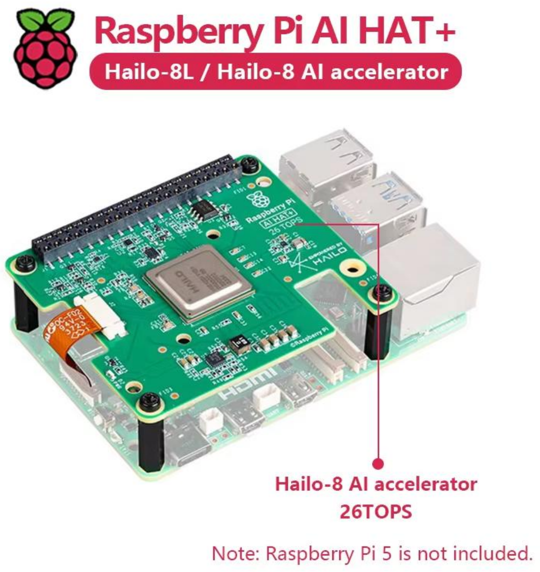
→ **This presented an opportunity**

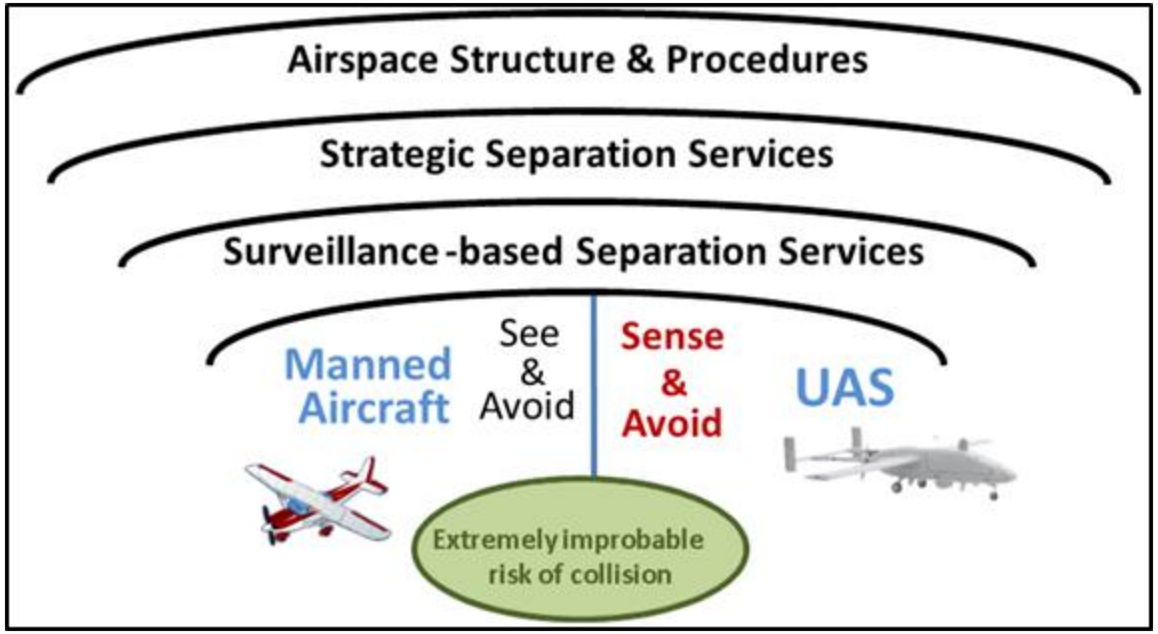
- ALQAYSI, ET.AL., "*A TEMPORAL BOOSTED YOLO-BASED MODEL FOR BIRDS DETECTION AROUND WIND FARMS*", JOURNAL OF IMAGING, 7, 227, 2021.
- ARRIOLA-VALVERDE, ET.AL., "*COFFEE CROP DETECTION FROM UAS ORTHOMAPS WITH CONVOLUTIONAL NEURAL NETWORKS*", 2023 IEEE CONFERENCE ON AGRIFOOD ELECTRONICS, 2023.
- DI CECIO, ET.AL., "*ON-BOARD DRONE CLASSIFICATION WITH DEEP LEARNING AND SYSTEM-ON-CHIP IMPLEMENTATION*", J. PHYS.: CONF. SER. **2716** 012059, 2024.
- GALLAGHER, ET.AL., "*SURVEYING YOU ONLY LOOK ONCE (YOLO) MULTISPECTRAL OBJECT DETECTION ADVANCEMENTS, APPLICATIONS, AND CHALLENGES*", IEEE ACCESS, *DOI 10.1109/ACCESS.2025.3526458, 2024.*
- GERARDI, "*THE APPLICABILITY OF MACHINE LEARNING METHODS ON INFRARED VIDEO DATA*", U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND ARMAMENTS CENTER, 2020.
- LAWRENCE, ET.AL., "*UAS-BASED REAL-TIME DETECTION OF RED-COCKADED WOODPECKER CAVITIES IN HETEROGENEOUS LANDSCAPES USING YOLO OBJECT DETECTION ALGORITHMS*", REMOTE SENS. 15, 883, 2023.
- SCHERMANN, ET.AL.,. "RISK-AWARE INTRUSION DETECTION AND PREVENTION SYSTEM FOR AUTOMATED UAS", 2023 IEEE 34TH ISSREW, 2023.
- SUN, ET.AL., "*FBD-SV-2024: FLYING BIRD OBJECT DETECTION DATASET IN SURVEILLANCE VIDEO*", WWW.NATURE.COM/SCIENTIFICDATA/, *12:530* | HTTPS://DOI.ORG/10.1038/S41597-025-04872-6, 2025.
- TIN, ET.AL., "*EXPLORING AN ALTERNATIVE APPROACH: RCNN-SVM FOR WEED IDENTIFICATION IN AERIAL IMAGES CAPTURED BY UAS*", 2024 INTERNATIONAL CONFERENCE ON EMERGING TRENDS IN NETWORKS AND COMPUTER COMMUNICATIONS, 2024.
- ULLAH, ET.AL., "*AUTONOMOUS CONTROL WITH VISION AND DEEP LEARNING: A RASPBERRY PI EDGE COMPUTING PLATFORM FOR OBSTACLE DETECTION IN SUAV PATH*", 7TH INTERNATIONAL COMMNET CONFERENCE, 2024.

PROJECT OBJECTIVE

Goal: Implement real-time bird detection on Raspberry Pi 5 and compare performance with and without Hailo-8L.

Why: Birds are unpredictable collision hazards for sUAS; onboard detection could support collision avoidance.





WHAT WE HAVE ACCOMPLISHED

Hardware

- Raspberry Pi5 operational on two systems
- Hailo AI HAT recognized and working
- Camera input functioning

Software

- YOLO installed and running
- Bird-only class filter implemented
- FPS and temperature logging added

Documentation

- Paper draft in progress
- All preliminary sections written
- Testing protocol documented for reproducibility

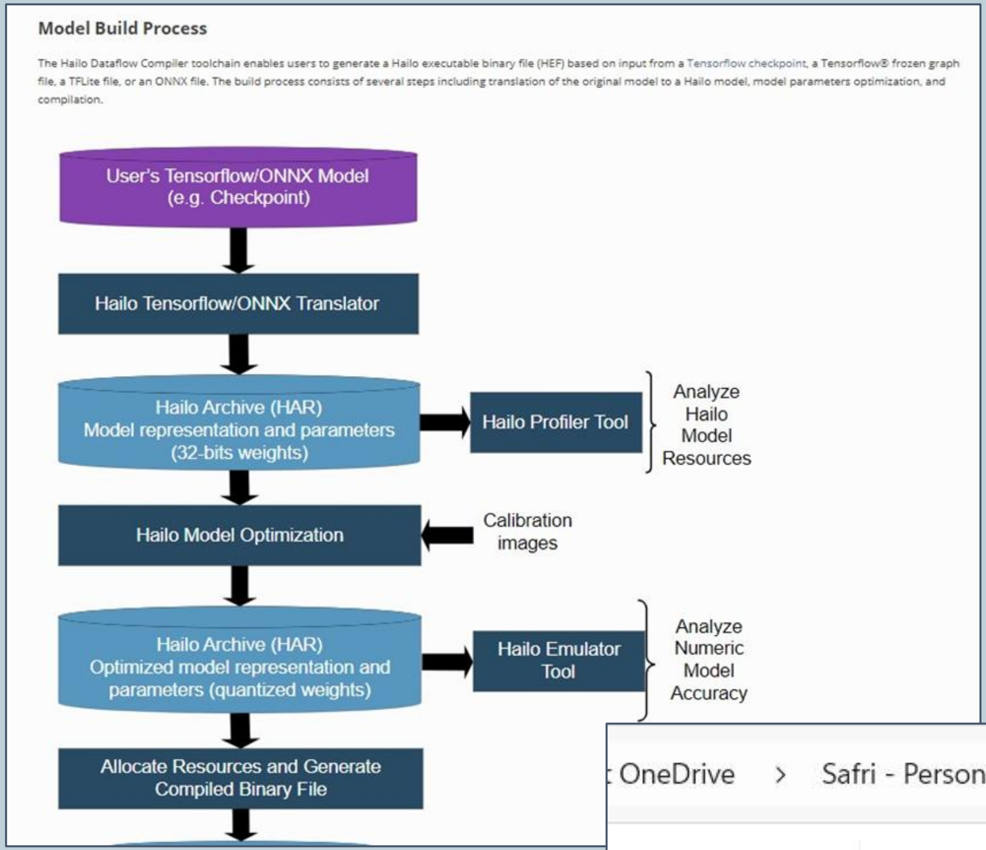
MODEL'S PROBLEM

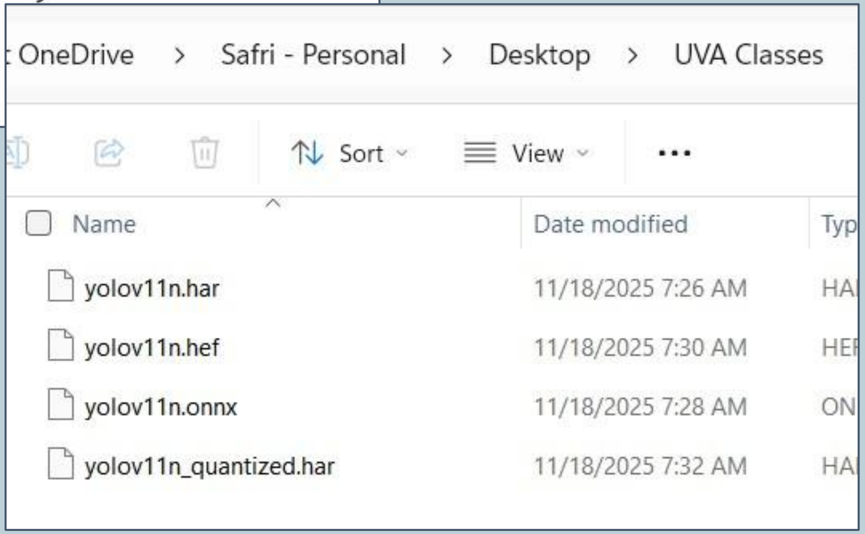
- Hailo 8L – Uses Proprietary .hef
- YOLO Model is .pt
- Hailo Dataflow Compiler cannot run on Pi5

Conversion Flow

.pt → ONNX → HAR → HAR QUANTIZED → .HEF

Hailo DFC Flow





HARDWARE VALIDATION

PI5 CPU ONLY = 6.25 FPS

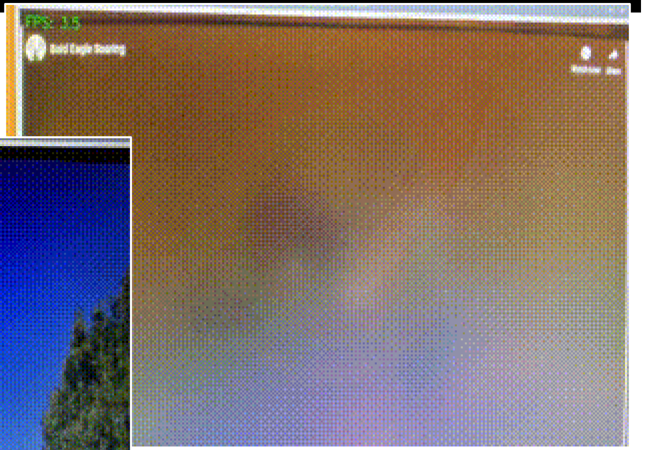
safri@raspberrypi: ~

File Edit Tabs Help

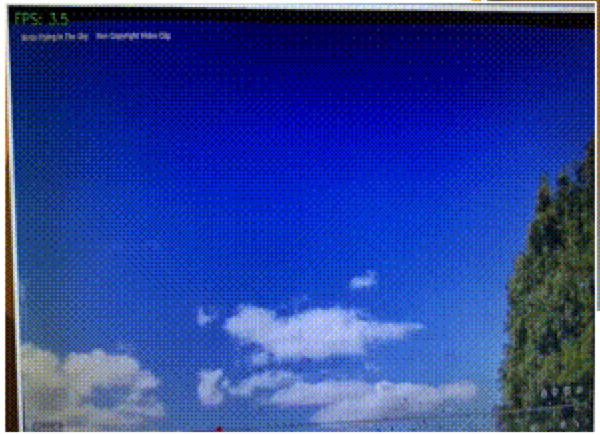
```
safri@raspberrypi:~ $ source ~/yolo_cpu_venv/bin/activate
(yolo_cpu_venv) safri@raspberrypi:~ $ python3 yolo_cpu_benchmark.py
2025-11-16 15:28:09.675766783 [W:onnxruntime:Default, device_discovery.cc
U device discovery failed: device_discovery.cc:89 ReadFileContents Failed
1/device/vendor"
Loading YOLOv8n ONNX model on CPU...
Input name: images
Input shape: [1, 3, 640, 640]

YOLOv8n CPU-only FPS on Raspberry Pi 5: 6.25 FPS

(yolo_cpu_venv) safri@raspberrypi:~ $
```



FPS: 3.5



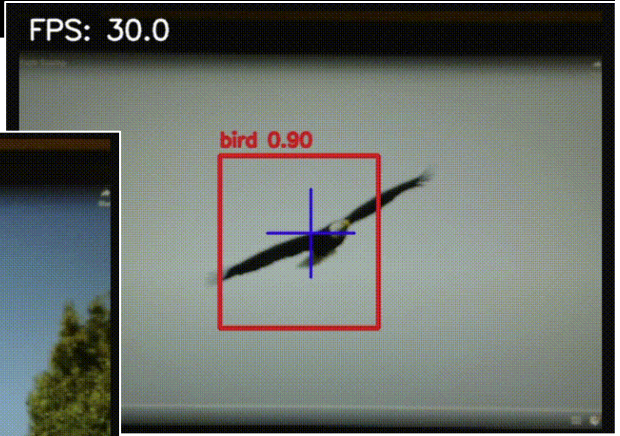
PI5 + AI HAT = 58.67 FPS

safri@raspberrypi: ~

File Edit Tabs Help

```
safri@raspberrypi:~ $ hailortcli benchmark ~/Downloads/yolov8s_h8l.hef
Starting Measurements...
Measuring FPS in HW-only mode
Network yolov8s/yolov8s: 100% | 882 | FPS: 58.67 | ETA: 00:00:00
Measuring FPS (and Power on supported platforms) in streaming mode
Network yolov8s/yolov8s: 100% | 882 | FPS: 58.67 | ETA: 00:00:00
Measuring HW Latency
Network yolov8s/yolov8s: 100% | 881 | HW Latency: 12.92 ms | ETA: 00:00:00

=======
Summary
=======
FPS       (hw_only)                 = 58.6732
          (streaming)               = 58.6738
Latency (hw)                        = 12.9192 ms
safri@raspberrypi:~ $
```



CURRENT STATUS: TESTING

Table 2 — Still Image Accuracy Summary

Tester Name	Platform	Total Samples	TP	FN	FP	TN	Precision	Recall	Notes
Jordan	Pi5	20	9	1	0	10	1	0.9	Failed to detect the partially occluded bird

Four Tests:

1. Still-Image Accuracy
2. Video-based Detection Performance
3. Robustness Under Varying Conditions
4. Short-Term Stability and Responsiveness

NEXT STEPS

1. Finish Testing

- Run full test on Pi-only vs Pi+Hailo

2. Analysis

- Compare performance between platforms
- Summarize results in tables
- Include in final report

3. Final Deliverables

- Baseline results
- Final write-up
- Presentation

Status: We expect to complete all deliverables on time.

THANK YOU

Jordan Capelle, Karl Garman, Mohomad Sabur